

Mapping from Prony to PRF parameters

We have developed a strategy, or recipe, for PRF nonlinear viscoelastic model calibration that begins with a Prony calibration and uses the Prony parameters to guide an initial guess at some of the PRF parameters. We took this approach because we have both a good intuitive feel for Prony parameters and good tools to calculate Prony parameters from test data. This approach also serves to shed some light on the differences between the linear viscoelastic behavior (Prony) and the nonlinear viscoelastic behavior (PRF). In order to facilitate this mapping we use a degenerate PRF model, meaning that we set $n=1$ and $m=0$ to simplify (degenerate) the PRF model to linear viscoelasticity.

Prony material model

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*Hyperelastic, Neo Hooke, Moduli=instant
550.0, 0.0
*Viscoelastic, Time=Prony
** g_i, kappa_i, tau_i
0.210, 0, 0.4
0.160, 0, 10.0
0.100, 0, 200.0
**
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degenerate PRF material model

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*Hyperelastic, NeoHooke, Moduli=instant
550.0, 0.0
*Viscoelastic, Nonlinear, NetworkId=1, SRatio=0.21, Law=strain
3.6075e-03, 1.0, 0.0
**
*Viscoelastic, Nonlinear, NetworkId=2, SRatio=0.16, Law=strain
1.8939e-04, 1.0, 0.0
**
*Viscoelastic, Nonlinear, NetworkId=3, SRatio=0.10, Law=strain
1.5152e-05, 1.0, 0.0
```

We assume that the material is a solid elastomer or polymer type material in which the bulk modulus, K , is infinite (material is incompressible) and the bulk behavior does not change with time. Thus we think of the Hyperelastic part of the material model as stipulating the shear modulus, G , of the material.

Further assumptions are:

1. The material is linear elastic.
2. The shear modulus is equal to the instantaneous shear modulus (for example, for the Neo Hookean material, we have $G=2 \cdot C_{10}$). Of course, if a better estimate of the shear modulus is available at the strain range of interest it should be used.
3. The viscous behavior of the PRF model is given by Law=strain.

The context, or scope, of this exercise is a PRF model that is limited to viscoelasticity – no plasticity, no Mullins effect, no TRS. Within this scope, the mechanical analog of the Prony model and the PRF model looks the same, as shown in Figure 1. The only difference is that the dashpot in the Prony series represents linear viscous behavior and in the PRF several nonlinear mathematical forms of viscous (creep) behavior are allowed. For this exercise we focus on the form called out by Law=strain, which is called the Power-Law Strain Hardening model (sometimes referred to as the Bailey-Norton power law creep model). We further degenerate this model by setting $n=1$ and $m=0$.

Degenerate (linearized) creep law

$$\dot{\epsilon}^{cr} = Aq \quad (1)$$

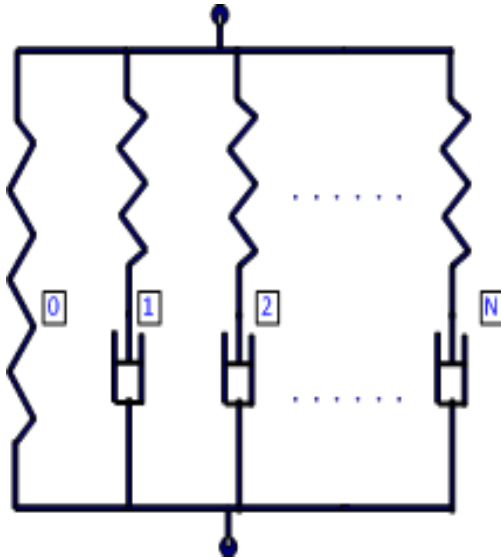


Figure 1, Mechanical Analog

For both models, the *Hyperelastic keyword is used to define the total elasticity. We think of the Prony series as a specification of g_i and τ_i terms ($k_i = 0$). The g_i terms are fractions that apportion the elastic behavior into each of the networks. Again, for this exercise, we think of the hyperelastic specification as giving us the total shear modulus, G . Thus, the shear modulus in network 1 is given by $G * g_1$. For the PRF model, the parameter SRATIO (stiffness ratio) serves exactly the same function as the g_i terms in the Prony series. So, part of our conversion from Prony parameters to degenerate PRF model parameters is:

$$SRATIO_i = g_i$$

The only other degenerate PRF parameter we need to define is the A term for each network.

Relaxation time for the degenerate PRF material.

Basic equations (1D case), written for just one of the viscous networks

Degenerate (linearized) creep law $\dot{\varepsilon}^{cr} = Aq$ (1)

Flow rule: $\dot{\varepsilon}^{cr} = \frac{3}{2q} \dot{\varepsilon}^{cr} \bar{\sigma}$ (2)

Stress-strain relation: $\bar{\sigma} = 2Gg(\varepsilon - \varepsilon^{cr})$ (3)

Combining the above equations produces:

$$\dot{\varepsilon}^{cr} = 3AGg(\varepsilon - \varepsilon^{cr}) \quad (4)$$

Or, rearranging slightly:

$$\dot{\varepsilon}^{cr} + 3AGg\varepsilon^{cr} = 3AGg\varepsilon \quad (5)$$

Assuming $\varepsilon(t) = \varepsilon_0 = \text{constant}$ (stress relaxation loading) and solving (5) we obtain:

$$\varepsilon^{cr}(t) = \varepsilon_0 \left(1 - e^{-(3AGgt)}\right) \quad (6)$$

We recognize eq. 6 as the classical Prony exponential decay formula, where:

$$3AGg = \frac{1}{\tau} \quad (7)$$

Rearranging: $A = \frac{1}{3Gg\tau}$ (8)

If Neo-Hookean: $A = \frac{1}{6C_{10}g\tau}$ (9)